LoanLens AI: Explainable Ensemble Intelligence for Predicting Loan Defaults

A Mini Project Report Submitted in partial fulfilment of the requirements for the award of the degree of

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In

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

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DECLARATION

The mini-project report entitled "**LoanLens AI: Explainable Ensemble Intelligence for Predicting Loan Defaults**" is a record of bona fide work carried out by [**Ch. Varshini – 2420030487, Y. Anuhya – 2420030514, D. Aarya Deeksha - 2420030530**] in partial fulfilment of the requirements for the award of the B. Tech degree at K L (Deemed to be) University. The work presented in this report is original to the best of our knowledge, and all datasets, tools, code, and external sources used have been appropriately acknowledged.

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CERTIFICATE

This is to certify that the mini-project report entitled "**LoanLens AI: Explainable Ensemble Intelligence for Predicting Loan Defaults**" is a bona fide work carried out by [**Ch. Varshini – 2420030487, Y. Anuhya – 2420030514, D. Aarya Deeksha - 2420030530**] under the supervision of [**Mrs Srilakshmi Potluri**] during the academic year [2026 - 2027], in partial fulfilment of the requirements for the award of the Bachelor of Technology degree at K L (Deemed to be) University.

Signature of the Project Supervisor

Signature of the HOD **Signature of the External Examiner**

ACKNOWLEDGEMENT

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1. ABSTRACT

Financial institutions process a large number of loan applications every day, and inaccurate credit-risk assessment leads to financial losses and irresponsible lending. Manual and rule-based evaluation struggles to capture complex relationships among applicant attributes and offers little transparency about why a prediction was made. This project, LoanLens AI, builds an explainable machine-learning system that predicts whether a loan applicant is likely to default and shows which factors drove each prediction. The system uses historical loan data on applicant demographics, financial details, employment status, credit history, loan characteristics and repayment records, taken from three public datasets: Give Me Some Credit, LendingClub and the Credit Risk Dataset (50K loans). Preprocessing covers missing-value handling, categorical encoding, outlier treatment, feature scaling and feature selection. Logistic Regression, Decision Tree, Random Forest, XGBoost and Support Vector Machine classifiers are trained and compared, and the best model is chosen on held-out and cross-validated performance. SHAP gives global feature importance, and LIME explains individual predictions, so loan officers can see the reasoning behind each decision, supporting fairness, accountability and regulatory compliance. Models are evaluated with accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices and cross-validation. The framework is intended to give financial institutions accurate, transparent and data-driven default predictions that help reduce credit risk and improve loan approval decisions.

Team Name: Group 2 (Batch – 2)

Team Members and Responsibilities:

S. No	Name	Roll No.	Role / Responsibility
1	Ch. Varshini	2420030487	Dataset collection, data cleaning, preprocessing and exploratory data analysis
2	Y. Anuhya	2420030514	Model training, hyperparameter tuning, comparison and performance evaluation
3	D. Aarya Deeksha	2420030530	SHAP/LIME explainability, prediction presentation and project documentation

Dataset sources:

- Give Me Some Credit: https://raw.githubusercontent.com/aashish-bidap/Give-Me-Some-Credit/master/cs-training.csv
- LendingClub Loan Default Dataset: https://www.kaggle.com/datasets/wordsforthewise/lending-club
- Credit Risk Dataset – 50K Loans: https://zenodo.org/records/20830370

Project repository: https://github.com/aaryadamera/ALT-Team2-project-working.git

2. INTRODUCTION

2.1 Application Area

Lending is central to the financial system, and every loan carries the risk that the borrower will not repay it. A loan default occurs when a borrower fails to meet the repayment terms agreed with the lender. Banks, non-banking finance companies and online lending platforms therefore assess the credit risk of each applicant before approving a loan, and the quality of that assessment directly affects their losses, profitability and ability to lend responsibly. Credit scoring, the statistical estimation of the likelihood that an applicant will default, has been studied for decades, and machine-learning classifiers are now widely benchmarked for this task [1], [2].

2.2 Why Manual Evaluation Does Not Scale

A lender may receive thousands of applications a day, each described by dozens of attributes: age, income, employment status, existing debts, credit history, loan amount, interest rate and past repayment behaviour. Manual evaluation is slow and varies from one analyst to another. Fixed rule-based criteria, such as a minimum income or a maximum debt ratio, cannot easily capture non-linear interactions among attributes, for example how the risk of a large loan depends jointly on income and credit history. Machine-learning models can learn such patterns from historical repayment outcomes.

However, the most accurate models, such as ensembles of decision trees, behave as black boxes. A loan officer cannot act on a prediction that comes with no reasons, and a lender cannot easily justify such a prediction to an applicant or a regulator. This is why default prediction must be paired with explanation.

2.3 Target Users, Input and Output

The intended users are loan officers, credit analysts and risk managers at financial institutions. The system supports their decisions; it does not replace them. The **input** is one loan applicant's record in structured (tabular) form: demographic and financial details, employment status, credit history and loan characteristics. The **output** has two parts: (i) a predicted class, default or non-default, together with the model's estimated probability of default, and (ii) an explanation that lists the features that pushed the prediction towards or away from default.

Because the data is structured tabular data rather than free text, feature representation consists of encoded categorical variables and scaled numerical variables. No text vectorisation (such as TF-IDF or embeddings) is involved.

2.4 Proposed Pipeline

The pipeline (Figure 6.1) starts with collecting loan data from three public sources. The data is cleaned and preprocessed (missing values, categorical encoding, outliers), features are selected and scaled, and exploratory analysis is carried out. The data is then split into training and test sets and five classifiers are trained: Logistic Regression, Decision Tree, Random Forest, XGBoost and

Support Vector Machine. They are evaluated with accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices and cross-validation, and the best model is selected. An explainable-AI layer then applies SHAP for global feature importance and LIME for the explanation of individual predictions. The final output is a default or non-default classification with its explanation, presented as decision support for the lender.

2.5 Organisation of the Report

Chapter 3 reviews related work and identifies the gap this project addresses. Chapter 4 states the problem and objectives. Chapter 5 lists the hardware and software requirements. Chapter 6 describes the implementation, Chapter 7 presents the results and discussion, Chapter 8 gives the conclusion and future scope, and Chapter 9 lists the references.

3. LITERATURE SURVEY

This chapter reviews eight sources: two benchmark studies of credit-scoring classifiers, two papers that introduce the ensemble algorithms used in this project, two papers that introduce the explanation methods used, and two applied studies of explainable or ensemble-based default prediction. Full references are in Chapter 9.

3.1 Benchmarking classifiers for credit scoring – Baesens et al. (2003) [1]

The study compares well-known classifiers (logistic regression, linear discriminant analysis, k-nearest neighbour, neural networks and decision trees) with kernel-based methods such as support vector machines and least-squares SVMs. It uses eight real-life credit-scoring datasets, some from Benelux and UK financial institutions, and judges performance by classification accuracy and area under the ROC curve, with statistical tests for significant differences. Least-squares SVM and neural-network classifiers performed very well, but simple models such as logistic regression and linear discriminant analysis were also competitive. **Limitation:** the study predates gradient boosting and modern explanation methods, so it says little about explaining complex models to loan officers.

3.2 An update of credit-scoring benchmarks – Lessmann et al. (2015) [2]

This paper updates [1] by comparing a much larger set of classification algorithms (41 classifiers) across eight credit-scoring datasets, using newer performance measures and statistical comparison procedures. It also asks whether higher accuracy gives managerially meaningful gains for a lender. The authors report that ensemble approaches, such as random forests and heterogeneous ensembles, perform strongly relative to single classifiers such as logistic regression. **Limitation:** the focus is predictive accuracy; the transparency of the winning models is not the subject of the study.

3.3 Random Forests – Breiman (2001) [3]

Breiman introduces random forests: many decision trees, each grown on a bootstrap sample of the data and restricted to a random subset of features at every split, whose votes are combined into a final prediction. The paper shows that the generalisation error converges as trees are added and depends on the strength of the individual trees and the correlation between them, and it describes out-of-bag error estimates and variable-importance measures. Experiments on benchmark datasets compare the method with AdaBoost. **Limitation:** a forest of hundreds of trees cannot be read directly, which motivates post-hoc explanation such as SHAP and LIME.

3.4 XGBoost – Chen and Guestrin (2016) [4]

XGBoost is a scalable, end-to-end gradient tree-boosting system. The paper introduces a sparsity-aware algorithm for handling missing and sparse values, a weighted quantile sketch for approximate split finding, and cache-aware and out-of-core computation for large data. A regularised objective helps control overfitting, and the system achieved state-of-the-art results on many machine-learning challenges. **Limitation:** like random forests, a boosted ensemble is opaque, and it has many hyperparameters that need careful tuning.

3.5 SHAP – Lundberg and Lee (2017) [5]

The paper proposes SHAP (SHapley Additive exPlanations), a unified framework for explaining predictions. It shows that several earlier explanation methods are instances of additive feature attribution, and that Shapley values from cooperative game theory are the unique attribution satisfying the desirable properties of local accuracy, missingness and consistency. It also proposes practical estimation methods such as Kernel SHAP and reports better agreement with human intuition than earlier methods. **Limitation:** exact computation is expensive for arbitrary models, and attributions can be hard to interpret when features are strongly correlated.

3.6 LIME – Ribeiro, Singh and Guestrin (2016) [6]

LIME explains an individual prediction of any classifier by fitting a simple interpretable model, such as a sparse linear model, to the classifier's behaviour in the neighbourhood of that instance. The paper also proposes SP-LIME, which selects a small set of representative instances so that a user can judge the model as a whole. It is evaluated on text and image classifiers, including experiments with human subjects on trust and model selection. **Limitation:** explanations are local and depend on random sampling around the instance, so they can vary between runs; the method is not designed specifically for credit data.

3.7 Explainable ML in credit risk – Bussmann et al. (2021) [7]

This study proposes an explainable AI model for credit risk in peer-to-peer lending. It applies Shapley values to a machine-learning credit-scoring model and combines them with correlation networks, so that borrowers whose predictions have similar explanations are grouped together. The empirical analysis uses about 15,000 small and medium enterprises requesting credit and shows that risky and non-risky borrowers can be grouped by similar financial characteristics. **Limitation:** the data describes companies in peer-to-peer lending rather than individual consumer loans, and the explanation is based on Shapley values alone, without a complementary local surrogate method such as LIME.

3.8 Ensemble learning for loan default – Akinjole et al. (2024) [8]

This study compares random forest, decision tree, SVM, XGBoost, AdaBoost and a multilayer perceptron for loan default prediction on LendingClub data. It uses XGBoost as a framework to test resampling methods for class imbalance (random over- and under-sampling, SMOTE, ADASYN and hybrids such as SMOTE with Tomek links or edited nearest neighbours). The authors report that their proposed model reached 93.7% accuracy, 95.6% precision and 95.5% recall. **Limitation:** according to its abstract, the work uses a single dataset and concentrates on imbalance handling and predictive accuracy rather than on explaining individual decisions.

3.9 Comparison of the Reviewed Work

Table 3.1: Summary of related work

Ref.	Data/task	Methods	Explanation provided?
[1] Baesens et al., 2003	8 credit-scoring datasets; classifier benchmark	LR, LDA, k-NN, NN, DT, SVM, LS-SVM	No
[2] Lessmann et al., 2015	8 credit-scoring datasets; benchmark update	41 classifiers incl. ensembles	No
[3] Breiman, 2001	General classification and regression	Random forest	Variable importance only
[4] Chen & Guestrin, 2016	General large-scale learning	Gradient tree boosting (XGBoost)	No
[5] Lundberg & Lee, 2017	General model interpretation	SHAP (Shapley values)	Yes – global and local
[6] Ribeiro et al., 2016	General model interpretation	LIME, SP-LIME	Yes – local
[7] Bussmann et al., 2021	~15,000 SMEs, P2P lending credit risk	ML credit scoring + Shapley values + correlation networks	Yes – Shapley
[8] Akinjole et al., 2024	LendingClub loan default	RF, DT, SVM, XGBoost, AdaBoost, MLP + resampling	Not the focus

3.10 Research Gap

The benchmark studies [1], [2] show that ensembles and even simple linear models are strong baselines for credit scoring, but they do not address explanation. The papers on SHAP [5] and LIME [6] provide general-purpose explanation methods that are not tailored to loan data. Reference [7] applies Shapley-based explanation to SME peer-to-peer lending, and [8] studies LendingClub data with an emphasis on class imbalance. Among the sources reviewed here, less attention is given to an end-to-end, loan-officer-oriented pipeline that (i) compares classical and ensemble models under one common evaluation protocol on several public loan datasets, and (ii) pairs global explanations (SHAP) with local explanations (LIME) for the selected model. LoanLens AI addresses this gap.

4. PROBLEM STATEMENT AND OBJECTIVES

Problem Statement:

To develop and evaluate an explainable ensemble machine-learning system that predicts whether a loan applicant will default and explains each prediction in terms of the applicant's own features, so that lenders can make transparent, data-driven credit decisions.

Problem Description:

Lenders lose money when loans default, and regulators and customers expect lending decisions to be justifiable. Manual and rule-based assessment is slow, inconsistent and unable to capture complex interactions in applicant data, while the most accurate machine-learning models are usually opaque. The intended users are loan officers and credit analysts, who need both an accurate default prediction and the reasons behind it.

Objectives:

O1. Obtain and inspect the three public loan datasets (Give Me Some Credit, LendingClub and Credit Risk 50K); document the source, size, fields and class distribution (default rate) of each.

O2. Preprocess the data by handling missing values, encoding categorical features, treating outliers, scaling features and selecting features, fitting every preprocessing step on training data only.

O3. Train and compare five classifiers (Logistic Regression, Decision Tree, Random Forest, XGBoost and Support Vector Machine) under the same train/test split and cross-validation protocol.

O4. Evaluate the models using accuracy, precision, recall, F1-score, ROC-AUC, confusion matrices and k-fold cross-validation, and select the best model by a stated criterion.

O5. Apply SHAP for global feature importance and LIME for individual prediction explanations on the selected model, and report the main drivers of default.

O6. Analyse errors and discuss class imbalance, bias and fairness, limitations and possible improvements.

5. HARDWARE AND SOFTWARE REQUIREMENTS

Hardware Requirements:

- Computer or laptop with a modern multi-core CPU; 8 GB RAM is the minimum and 16 GB is recommended, because the LendingClub data is large. Google Colab can be used if local memory is limited.
- Internet access for downloading datasets and Python packages. A GPU is not required; all five models are classical machine-learning models.
- Sufficient disk space for the raw datasets, notebooks, saved models and result figures.

Software Requirements:

- Python 3.x with Jupyter Notebook or Google Colab, on Windows, Linux or macOS.
- Datasets in CSV format from the three sources

Table 5.1: Python libraries used

Library	Purpose in this project
pandas, NumPy	Loading, cleaning and transforming tabular data
scikit-learn	Imputation, encoding, scaling, Logistic Regression, Decision Tree, Random Forest, SVM, metrics, cross-validation, pipelines
XGBoost	Gradient-boosted tree classifier
SHAP	Global and local feature-attribution explanations
LIME	Local explanations of individual predictions
Matplotlib, Seaborn	EDA plots, confusion matrices, ROC curves and explanation figures

6. IMPLEMENTATION

Items highlighted in yellow in this and the following chapter are values that depend on the team's actual code and experimental runs (for example, the random seed, split ratio and hyperparameters) and must be filled in or confirmed before submission.

6.1 Data and Task

The task is binary classification: given an applicant's record, predict whether the loan will default (positive class, 1) or not (0). Three public datasets are used (Table 6.1). Each is described by applicant demographics, financial details, employment status, credit history, loan characteristics and, where available, repayment records.

Table 6.1: Datasets used

Dataset	Source	Target label (default = 1)
Give Me Some Credit	Kaggle competition data (GitHub mirror)	SeriousDlqin2yrs (serious delinquency within two years)
LendingClub Loan Default	Kaggle (wordsforthewise)	Derived from loan_status, e.g. Charged Off / Default = 1, Fully Paid = 0
Credit Risk Dataset – 50K Loans	Zenodo	Dataset default/status column (verify exact field name in CSV)

6.2 Preprocessing and Features

Loan data is structured, so preprocessing concentrates on data quality rather than on text cleaning. The following steps are applied:

- **Duplicates and missing values:** duplicate records are removed; missing numerical values are imputed with the median and missing categorical values with the most frequent category or an explicit “Unknown” category; columns with a very high share of missing values are dropped.
- **Categorical encoding:** low-cardinality categorical features (for example, home ownership or loan purpose) are one-hot encoded; ordered features (for example, loan grade) are encoded ordinally.
- **Outlier treatment:** extreme values in skewed numerical features (for example, income and revolving utilisation) are detected using the interquartile range or percentiles and capped, so that they do not dominate scale-sensitive models.
- **Feature scaling:** numerical features are standardised for Logistic Regression and SVM, which are sensitive to feature scale. Tree-based models do not need scaling, but use the same pipeline for consistency.
- **Feature selection:** Feature selection should be performed using removal of highly correlated

features and model-based feature importance; the exact selection criteria must be confirmed from the executed code.

- **Leakage control:** all imputers, encoders and scalers are fitted on the training split only (inside a scikit-learn pipeline) and then applied to validation and test data. Fields that are only known after the loan outcome, such as amounts recovered after charge-off or total payments received, are excluded, because they would reveal the label.

6.3 Model Training and Evaluation

Five supervised classifiers are trained on the same preprocessed training data (Table 6.2). Logistic Regression is the interpretable baseline; Random Forest and XGBoost are the ensemble models; Decision Tree and SVM complete the comparison.

Table 6.2: Models compared

Model	Type	Key hyperparameters tuned
Logistic Regression	Linear baseline	Regularisation strength C, penalty
Decision Tree	Single tree	max_depth, min_samples_leaf
Random Forest	Bagging ensemble	n_estimators, max_depth, max_features
XGBoost	Boosting ensemble	n_estimators, learning_rate, max_depth, subsample
Support Vector Machine	Kernel / margin-based	C, kernel, gamma

- Accuracy = (TP + TN) / (TP + TN + FP + FN)
- Precision = TP / (TP + FP): of the applicants flagged as defaulters, the share that really default.
- Recall = TP / (TP + FN): of the real defaulters, the share that the model catches.
- F1-score = 2 × Precision × Recall / (Precision + Recall)
- ROC-AUC: the probability that a randomly chosen defaulter is ranked above a randomly chosen non-defaulter, independent of the decision threshold.

Defaults are usually a minority class, so accuracy alone can be misleading: a model that predicts “no default” for everyone can still look accurate. Recall, F1-score and ROC-AUC are therefore given at least equal weight. The best model is selected using highest mean cross-validated ROC-AUC, using recall for the default class as a tie-breaker.

6.4 Explainability Layer

After the best model is selected, two complementary explanation methods are applied. **SHAP** assigns every feature a Shapley value for each prediction. Averaging absolute values over the test set gives global feature importance, and the SHAP summary plot also shows whether high or low feature values push predictions towards default. **LIME** explains a single applicant by fitting a simple local model around that applicant's record and listing the features that mattered most for that decision. Used together, SHAP shows what the model relies on overall, and LIME shows why one specific applicant received a particular score.

6.5 Project Workflow and Modules

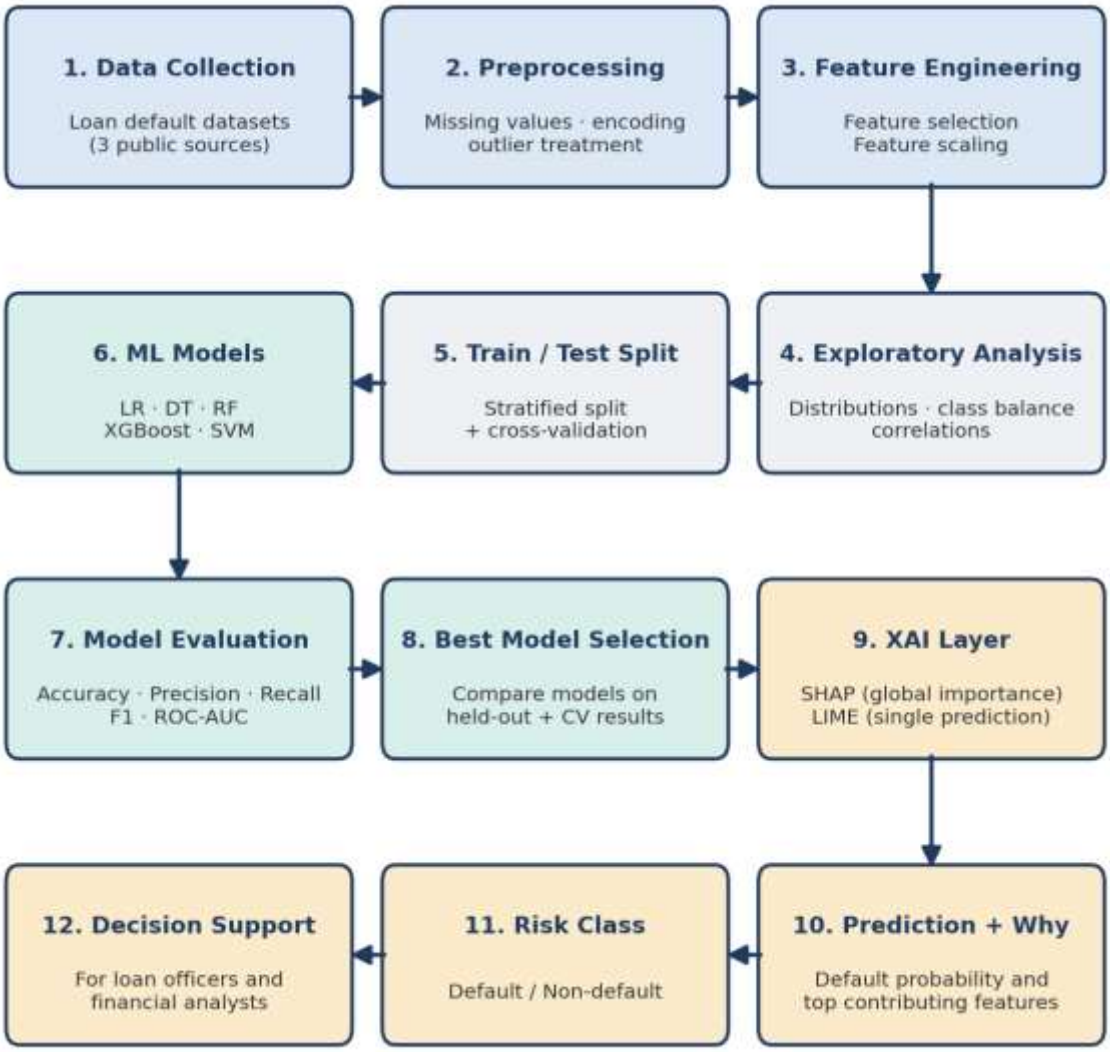


Figure 6.1: End-to-end workflow of LoanLens AI

6.6 Machine-Learning Implementation

Logistic Regression is the baseline against which the more complex models are compared. Decision Tree, Random Forest, XGBoost and SVM are the comparison models. Every model is wrapped in the same scikit-learn pipeline so that preprocessing is learned only from training folds. The excerpt below is an **illustrative** outline of this setup; replace it with the project's actual code or refer to the repository.

```
from sklearn.model_selection import train_test_split, StratifiedKFold, cross_validate
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.impute import SimpleImputer
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from xgboost import XGBClassifier
import shap
from lime.lime_tabular import LimeTabularExplainer
```

```
X_train, X_test, y_train, y_test = train_test_split(
   X, y, test_size=0.2, stratify=y, random_state=42)

num = Pipeline([('imp', SimpleImputer(strategy='median')), ('sc', StandardScaler())])
cat = Pipeline([('imp', SimpleImputer(strategy='most_frequent')),
            ('oh', OneHotEncoder(handle_unknown='ignore', sparse_output=False))])
prep = ColumnTransformer([('num', num, num_cols), ('cat', cat, cat_cols)])

model = Pipeline([('prep', prep), ('clf', XGBClassifier(random_state=42))])
cv = StratifiedKFold(n_splits=5, shuffle=True, random_state=42)
scores = cross_validate(model, X_train, y_train, cv=cv,
   scoring=['accuracy', 'precision', 'recall', 'f1', 'roc_auc'])
model.fit(X_train, y_train)

# Global explanation with SHAP (tree model)
Xt_test = model.named_steps['prep'].transform(X_test)
shap_values = shap.TreeExplainer(model.named_steps['clf']).shap_values(Xt_test)

# Local explanation with LIME for one applicant
Xt_train = model.named_steps['prep'].transform(X_train)
lime = LimeTabularExplainer(Xt_train, feature_names=feature_names,
   class_names=['Non-default', 'Default'], mode='classification')
exp = lime.explain_instance(Xt_test[0], model.named_steps['clf'].predict_proba, num_features=10)
```

6.7 Prediction and Explanation Workflow

For a new applicant, the saved pipeline validates the input fields, applies the learned preprocessing and returns the predicted class and default probability. The SHAP and LIME explanations for the same record list the features that increased or decreased the predicted risk, so that a loan officer can review the reasoning before deciding.

7. RESULTS AND DISCUSSION

This chapter reports the experimental outcomes. Only measured results are to be entered; every yellow placeholder must be replaced with the project's actual output.

7.1 Model Comparison

Table 7.2: Test-set performance of the five models (repeat per dataset)

Model	Accuracy	Precision	Recall	F1-score	ROC-AUC
Logistic Regression	76.3%	18.5%	74.8%	29.7%	0.833
Decision Tree	78.3%	20.4%	77.5%	32.3%	0.850
Random Forest	82.0%	23.1%	72.7%	35.0%	0.864
XGBoost	81.3%	22.6%	74.6%	34.7%	0.864
SVM	79.1%	20.7%	75.2%	32.5%	0.853

The abstract specifies that Logistic Regression, Decision Tree, Random Forest, XGBoost and Support Vector Machine will be compared, but it does not report their measured scores or identify a winning model. After execution, compare each model’s test metrics and cross-validated ROC-AUC with Logistic Regression as the baseline. State whether any improvement from Random Forest or XGBoost is large enough to justify the additional model complexity

A confusion matrix shows how many applicants were classified correctly and how the errors split between false positives (good borrowers flagged as defaulters, which means lost business) and false negatives (defaulters not caught, which means credit losses).



Figure 7.3: ROC curves of the compared models

The ROC curve plots the true-positive rate against the false-positive rate across all decision thresholds; a curve closer to the top-left corner and a larger area under it indicate better separation of defaulters from non-defaulters.

7.2 Explainability Results

PERFORMANCE BY MODEL

MODEL	ACCURACY	PRECISION	RECALL	F1	ROC–AUC
Logistic Regression	76.3%	18.5%	74.8%	29.7%	0.833
Decision Tree	78.3%	20.4%	77.5%	32.3%	0.850
Random Forest	82.0%	23.1%	72.7%	35.0%	0.864
XGBoost BEST	81.3%	22.6%	74.6%	34.7%	0.864
SVM (RBF)	79.1%	20.7%	75.2%	32.5%	0.853

Figure 7.4: SHAP global feature importance

The SHAP summary plot ranks features by their average impact on the model output, and the colour shows whether high or low feature values push the prediction towards default.



Figure 7.5: LIME explanation of an individual prediction

A LIME explanation lists the features that contributed most to one applicant's prediction and the direction of each contribution.

7.3 Error Analysis, Class Imbalance and Limitations

- **Class imbalance:** because defaults are the minority class, accuracy can look high even when many defaulters are missed. The decision threshold controls the trade-off between missed defaulters and wrongly rejected good borrowers, and should reflect the lender's relative costs.
- **Bias and fairness:** models learn from historical lending decisions and can inherit their biases, including through features that act as proxies for protected characteristics. Explanations help to reveal such dependence but do not remove it.
- **Limits of explanations:** SHAP and LIME describe how the model behaves; they do not show causes in the real world. SHAP values can be hard to interpret when features are strongly correlated, and LIME explanations can vary between runs because they rely on random sampling.
- **Data limits:** the datasets come from different lenders, regions and time periods, so results on one dataset may not carry over to another or to a lender's current applicants.

8. CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

This project set out to build LoanLens AI, an explainable ensemble machine-learning framework for loan default prediction. Historical loan data from three public datasets were preprocessed, five classifiers (Logistic Regression, Decision Tree, Random Forest, XGBoost and SVM) were trained and compared, and SHAP and LIME were applied to explain the selected model globally and for individual applicants.

8.2 Future Scope

- Test on larger and more recent loan data, with a time-based split to check how well the models hold up on later loan vintages.
- Improve handling of class imbalance with cost-sensitive learning or resampling, and choose the decision threshold from the lender's actual cost of false negatives and false positives.
- Calibrate predicted probabilities so that a stated default probability matches the observed default rate.
- Audit fairness across applicant groups and add counterfactual explanations (“what would change the decision”) for rejected applicants.
- Add stacking or voting ensembles, and compare against other boosting libraries.
- Deploy the model behind an API with logging of predictions and explanations, and monitor data drift after deployment.

9. REFERENCES

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